KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 12 · MATH

The Houston Apartment Build — Teacher Lesson Plan & Answer Key

Learning objectives: Students use area and length models to explain why fractions are equivalent, generate equivalent fractions by multiplying or dividing the numerator and denominator by the same number (n/n), simplify fractions in one step using the greatest common factor, and convert between improper fractions and mixed numbers. Students also solve two-step word problems, explain their reasoning, find mistakes in others' work, and correct punctuation, capitalization, and grammar in math sentences.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Amara, who is helping her aunt's crew build a six-story apartment building in Houston; student types first and last name so results are sent.
2. Slide 1 — Warm-Up: Week 11 Review (12 pts): GCF, prime/composite, LCM, and a missing factor. The GCF item previews today's simplifying. If a student misses the GCF item, list the factors of both numbers together on paper before moving on.
3. Slide 2 — Pencil Break (Pre): student completes the Pre-Lesson Warm-Up sheet off screen.
4. Slide 3 — Hook: building in Houston (largest city in Texas, the "Bayou City"); meet the crew and preview the fraction jobs.
5. Slide 4 — Picture It: three same-size hallways tiled with 2, 4, and 8 tiles show 1/2 = 2/4 = 4/8. Ask: "When the tiles got smaller, what happened to the number of tiles AND the number of blue tiles?"
6. Slide 5 — Teach: multiply or divide by n/n because n/n = 1. Worked examples 2/7 = 8/28 and 14/16 = 7/8. Stress the warning that adding the same number does not work (1/2 vs. 2/3).
7. Slide 6 — Practice: equivalent fractions (12 pts).
8. Slide 7 — Teach: simplest form with the GCF using Ms. Patel's 36-square floor plan (24/36 = 2/3). Compare the three small steps with the one GCF step, and point out 4 of 6 rows on the grid.
9. Slide 8 — Practice: simplifying, choosing the GCF, and recognizing simplest form (12 pts).
10. Slide 9 — Vocabulary flip cards: fraction, numerator, denominator, equivalent fractions, simplest form, GCF, improper fraction, mixed number.
11. Slide 10 — Vocabulary check (6 pts).
12. Slide 11 — Teach: improper fractions and mixed numbers on a pipe number line (11/4 = 2 3/4 by division with remainder; 2 4/5 = 14/5 by changing wholes to fifths). Have the student say "the denominator stays the same."
13. Slide 12 — Practice: mixed numbers and improper fractions (12 pts).
14. Slide 13 — Real World: lobby tiles, two-step: subtract, then simplify (5 pts).
15. Slide 14 — Real World: pipe lengths and rebar, two-step conversions (5 pts).
16. Slide 15 — Think It Through: "which explanation is correct?" and "find the mistake" (10 pts). Ask the student to say why each wrong choice is wrong.

17. Slide 16 — Fix the Sentence (8 pts): four sentences on quotation marks in dialogue, its vs. it's, commas in a series, and subject-verb agreement. Students keep the same words in the same order. The answer must match one accepted form exactly (straight or curly quotes both work; the comma before "and" is optional; "It is" for "It's" and "The plumber measures" are also accepted).
18. Slide 17 — Match-Up Round (10 pts): 5 clue-to-answer matches, 2 pts each on the first try.
19. Slide 18 — Speed Trail (bonus XP only).
20. Slide 19 — Looking Back reflection (bonus XP only).
21. Slide 20 — Boss Battle (8 pts): two-step problems on inspection day (fraction not finished, in simplest form; pipe length as a simplified mixed number).
22. Slide 21 — Pencil Break (Post): student completes the Post-Lesson Challenge sheet off screen.
23. Slide 22 — Results: score out of 100 sent to KingCretot Experience; 85+ shows the fraction-builder congratulations with confetti; below 85 shows encouragement, and Play Again gives the next question set.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: WEEK 11 REVIEW (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: GCF (Week 11): type the greatest common factor of 12 and 18. GCF of 12 and 18 =	**6**	Factors of 12: 1, 2, 3, 4, 6, 12. Factors of 18: 1, 2, 3, 6, 9, 18. The greatest factor they share is 6. Today you will use a GCF like this to simplify 12/18.
r2_s0	Set 1: Prime and composite (Week 11): which number is prime?	**23**	23 has only two factors, 1 and 23. 21 = 3 × 7, 25 = 5 × 5, and 27 = 3 × 9, so those are composite.
r3_s0	Set 1: LCM (Week 11): what is the least common multiple of 4 and 6?	**12**	Multiples of 4: 4, 8, 12. Multiples of 6: 6, 12. The first one they share is 12. (24 is a common multiple, but not the least one. 2 is the GCF.)
r4_s0	Set 1: Factor pairs (Week 11): type the missing factor. 8 × ? = 56	**7**	8 × 7 = 56, so 7 and 8 are a factor pair of 56.
r1_s1	Set 2: GCF (Week 11): type the greatest common factor of 16 and 24. GCF of 16 and 24 =	**8**	Factors of 16: 1, 2, 4, 8, 16. Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. The greatest factor they share is 8. Today you will use a GCF like this to simplify 16/24.
r2_s1	Set 2: Prime and composite (Week 11): which number is composite?	**33**	33 = 3 × 11, so it has more than two factors. 31, 37, and 29 each have only two factors, so they are prime.
r3_s1	Set 2: LCM (Week 11): what is the least common multiple of 3 and 4?	**12**	Multiples of 3: 3, 6, 9, 12. Multiples of 4: 4, 8, 12. The first one they share is 12. (24 is a common multiple, but not the least one. 6 is not a multiple of 4.)
r4_s1	Set 2: Factor pairs (Week 11): type the missing factor. 9 × ? = 54	**6**	9 × 6 = 54, so 6 and 9 are a factor pair of 54.

ID	Question	Answer	Why (explain to student)
r1_s2	Set 3: GCF (Week 11): type the greatest common factor of 20 and 30. GCF of 20 and 30 =	**10**	Factors of 20: 1, 2, 4, 5, 10, 20. Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. The greatest factor they share is 10. Today you will use a GCF like this to simplify 20/30.
r2_s2	Set 3: Prime and composite (Week 11): which number is prime?	**41**	41 has only two factors, 1 and 41. 39 = 3 × 13, 49 = 7 × 7, and 51 = 3 × 17, so those are composite.
r3_s2	Set 3: LCM (Week 11): what is the least common multiple of 6 and 8?	**24**	Multiples of 6: 6, 12, 18, 24. Multiples of 8: 8, 16, 24. The first one they share is 24. (48 is a common multiple, but not the least one. 2 is the GCF.)
r4_s2	Set 3: Factor pairs (Week 11): type the missing factor. 7 × ? = 63	**9**	7 × 9 = 63, so 7 and 9 are a factor pair of 63.

YOUR TURN: EQUIVALENT FRACTIONS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: Multiply the numerator and denominator by the same number. Type the missing numerator. 2/3 = ?/12	**8**	3 × 4 = 12, so multiply by 4/4. The numerator becomes 2 × 4 = 8. So 2/3 = 8/12.
e2_s0	Set 1: Which fraction is equivalent to 3/4?	**9/12**	Multiply by 3/3: 3 × 3 = 9 and 4 × 3 = 12, so 3/4 = 9/12. 6/12 is only 1/2, and 3/8 is smaller than 3/4.
e3_s0	Set 1: Divide the numerator and denominator by the same number. Which fraction equals 10/15?	**2/3**	Divide by 5/5: 10 ÷ 5 = 2 and 15 ÷ 5 = 3, so 10/15 = 2/3.
e1_s1	Set 2: Multiply the numerator and denominator by the same number. Type the missing numerator. 3/5 = ?/20	**12**	5 × 4 = 20, so multiply by 4/4. The numerator becomes 3 × 4 = 12. So 3/5 = 12/20.
e2_s1	Set 2: Which fraction is equivalent to 2/5?	**6/15**	Multiply by 3/3: 2 × 3 = 6 and 5 × 3 = 15, so 2/5 = 6/15. In 4/15 and 2/10, the numerator and denominator were not multiplied by the same number.
e3_s1	Set 2: Divide the numerator and denominator by the same number. Which fraction equals 12/16?	**3/4**	Divide by 4/4: 12 ÷ 4 = 3 and 16 ÷ 4 = 4, so 12/16 = 3/4.
e1_s2	Set 3: Multiply the numerator and denominator by the same number. Type the missing numerator. 5/6 = ?/18	**15**	6 × 3 = 18, so multiply by 3/3. The numerator becomes 5 × 3 = 15. So 5/6 = 15/18.
e2_s2	Set 3: Which fraction is equivalent to 4/7?	**8/14**	Multiply by 2/2: 4 × 2 = 8 and 7 × 2 = 14, so 4/7 = 8/14. In 8/7 and 4/14, only one part was doubled.
e3_s2	Set 3: Divide the numerator and denominator by the same number. Which fraction equals 14/21?	**2/3**	Divide by 7/7: 14 ÷ 7 = 2 and 21 ÷ 7 = 3, so 14/21 = 2/3.

YOUR TURN: SIMPLIFY IT (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Simplify using the GCF. Type the fraction in simplest form. 15/20 =	**3/4**	The GCF of 15 and 20 is 5. 15 ÷ 5 = 3 and 20 ÷ 5 = 4, so 15/20 = 3/4.
s2_s0	Set 1: Ms. Patel wants to simplify 16/40 in ONE step. What number should she divide the numerator and denominator by?	**8**	Factors of 16: 1, 2, 4, 8, 16. Factors of 40: 1, 2, 4, 5, 8, 10, 20, 40. The GCF is 8, so 16/40 ÷ 8/8 = 2/5 in one step. Dividing by 4 or 2 works, but takes more steps.
s3_s0	Set 1: Which fraction is already in simplest form?	**5/8**	5 and 8 have no common factor except 1, so 5/8 is in simplest form. 6/9 = 2/3, 4/10 = 2/5, and 8/12 = 2/3.
s1_s1	Set 2: Simplify using the GCF. Type the fraction in simplest form. 18/30 =	**3/5**	The GCF of 18 and 30 is 6. 18 ÷ 6 = 3 and 30 ÷ 6 = 5, so 18/30 = 3/5.
s2_s1	Set 2: Ms. Patel wants to simplify 27/36 in ONE step. What number should she divide the numerator and denominator by?	**9**	Factors of 27: 1, 3, 9, 27. Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. The GCF is 9, so 27/36 ÷ 9/9 = 3/4 in one step. 6 and 4 are not factors of 27.
s3_s1	Set 2: Which fraction is NOT in simplest form?	**9/15**	9 and 15 share the factor 3: 9/15 = 3/5. In 7/9, 5/12, and 8/11, the only common factor is 1.
s1_s2	Set 3: Simplify using the GCF. Type the fraction in simplest form. 20/28 =	**5/7**	The GCF of 20 and 28 is 4. 20 ÷ 4 = 5 and 28 ÷ 4 = 7, so 20/28 = 5/7.
s2_s2	Set 3: Ms. Patel wants to simplify 30/45 in ONE step. What number should she divide the numerator and denominator by?	**15**	Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. Factors of 45: 1, 3, 5, 9, 15, 45. The GCF is 15, so 30/45 ÷ 15/15 = 2/3 in one step. 9 is not a factor of 30.
s3_s2	Set 3: Which fraction is already in simplest form?	**7/10**	7 and 10 have no common factor except 1, so 7/10 is in simplest form. 6/14 = 3/7, 9/12 = 3/4, and 12/20 = 3/5.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: Fractions that name the same amount, like 1/2 and 2/4, are called ___.	**Equivalent fractions**	Fractions that name the same amount are equivalent fractions.
v2_s0	Set 1: Type your answer: the bottom number of a fraction, which tells how many equal parts make one whole, is the ___.	**denominator**	The denominator is the bottom number. In 3/4, the denominator 4 means the whole is cut into 4 equal parts.
v1_s1	Set 2: What do we call a fraction like 7/4, whose numerator is greater than its denominator?	**Improper fraction**	A fraction whose numerator is greater than or equal to its denominator is an improper fraction. 7/4 is more than one whole.
v2_s1	Set 2: Type your answer: the top number of a fraction, which tells how many parts you have, is the ___.	**numerator**	The numerator is the top number. In 3/4, the numerator 3 means you have 3 of the 4 equal parts.
v1_s2	Set 3: A fraction is in simplest form when the greatest common factor of its numerator and denominator is ___.	**1**	When the only common factor left is 1, you cannot divide any further, so the fraction is in simplest form.

ID	Question	Answer	Why (explain to student)
v2_s2	Set 3: Type your answer: a whole number and a fraction together, like 1 1/2, is called a mixed ___.	**number**	A whole number written with a fraction, like 1 1/2, is a mixed number.

YOUR TURN: MIXED NUMBERS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
m1_s0	Set 1: Which mixed number equals 13/4?	**3 1/4**	13 ÷ 4 = 3 R1. That is 3 wholes with 1 fourth left over, so 13/4 = 3 1/4.
m2_s0	Set 1: Change the mixed number to an improper fraction. Type the missing numerator. 2 2/3 = ?/3	**8**	2 wholes = 2 × 3 = 6 thirds. Add the 2 thirds: 6 + 2 = 8. So 2 2/3 = 8/3.
m3_s0	Set 1: Mr. Nguyen cuts a pipe that is 7/2 feet long. Which mixed number is the same length?	**3 1/2 feet**	7 ÷ 2 = 3 R1. That is 3 whole feet and 1 half foot, so 7/2 feet = 3 1/2 feet.
m1_s1	Set 2: Which mixed number equals 17/5?	**3 2/5**	17 ÷ 5 = 3 R2. That is 3 wholes with 2 fifths left over, so 17/5 = 3 2/5.
m2_s1	Set 2: Change the mixed number to an improper fraction. Type the missing numerator. 3 1/4 = ?/4	**13**	3 wholes = 3 × 4 = 12 fourths. Add the 1 fourth: 12 + 1 = 13. So 3 1/4 = 13/4.
m3_s1	Set 2: Mr. Nguyen cuts a pipe that is 11/3 feet long. Which mixed number is the same length?	**3 2/3 feet**	11 ÷ 3 = 3 R2. That is 3 whole feet and 2 thirds of a foot, so 11/3 feet = 3 2/3 feet.
m1_s2	Set 3: Which mixed number equals 19/6?	**3 1/6**	19 ÷ 6 = 3 R1. That is 3 wholes with 1 sixth left over, so 19/6 = 3 1/6.
m2_s2	Set 3: Change the mixed number to an improper fraction. Type the missing numerator. 4 3/5 = ?/5	**23**	4 wholes = 4 × 5 = 20 fifths. Add the 3 fifths: 20 + 3 = 23. So 4 3/5 = 23/5.
m3_s2	Set 3: Mr. Nguyen cuts a pipe that is 15/4 feet long. Which mixed number is the same length?	**3 3/4 feet**	15 ÷ 4 = 3 R3. That is 3 whole feet and 3 fourths of a foot, so 15/4 feet = 3 3/4 feet.

REAL WORLD: THE LOBBY TILES (5 POINTS)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Diego Ramírez, the tile setter, is covering a lobby floor with 24 square tiles. 18 of the tiles are blue, and the rest are white. Two steps: find how many tiles are white, then type the fraction of the floor that is white, in simplest form. Fraction that is white =	**1/4**	Step 1: 24 − 18 = 6 white tiles, so 6/24 of the floor is white. Step 2: the GCF of 6 and 24 is 6. 6 ÷ 6 = 1 and 24 ÷ 6 = 4, so the answer is 1/4.
w1_s1	Set 2: Diego Ramírez, the tile setter, is covering a laundry room floor with 40 square tiles. 15 of the tiles are gray, and the rest are white. Two steps: find how many tiles are white, then type the fraction of the floor that is white, in simplest form. Fraction that is white =	**5/8**	Step 1: 40 − 15 = 25 white tiles, so 25/40 of the floor is white. Step 2: the GCF of 25 and 40 is 5. 25 ÷ 5 = 5 and 40 ÷ 5 = 8, so the answer is 5/8.

ID	Question	Answer	Why (explain to student)
w1_s2	Set 3: Diego Ramírez, the tile setter, is covering a hallway floor with 32 square tiles. 20 of the tiles are blue, and the rest are white. Two steps: find how many tiles are white, then type the fraction of the floor that is white, in simplest form. Fraction that is white =	**3/8**	Step 1: 32 − 20 = 12 white tiles, so 12/32 of the floor is white. Step 2: the GCF of 12 and 32 is 4. 12 ÷ 4 = 3 and 32 ÷ 4 = 8, so the answer is 3/8.

REAL WORLD: PIPES AND REBAR (5 POINTS)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: Mr. Thanh Nguyen, the plumber, lines up 14 short pieces of pipe end to end. Each piece is 1/3 foot long. Two steps: write the total length as an improper fraction, then change it to a mixed number. How long is the line of pipe?	**4 2/3 feet**	Step 1: 14 pieces of 1/3 foot make 14/3 feet. Step 2: 14 ÷ 3 = 4 R2, so 14/3 = 4 2/3 feet.
w2_s1	Set 2: Darnell Brooks leads the rebar crew. The building plan says one steel bar must be 3 3/8 feet long, and the crew's cutting chart lists every length in eighths of a foot. Two steps: count the eighths in the whole feet, then add the extra eighths. Type the missing numerator. 3 3/8 feet = ?/8 feet	**27**	Step 1: 3 whole feet = 3 × 8 = 24 eighths. Step 2: 24 + 3 = 27, so 3 3/8 feet = 27/8 feet.
w2_s2	Set 3: Mr. Thanh Nguyen, the plumber, lines up 17 short pieces of pipe end to end. Each piece is 1/4 foot long. Two steps: write the total length as an improper fraction, then change it to a mixed number. How long is the line of pipe?	**4 1/4 feet**	Step 1: 17 pieces of 1/4 foot make 17/4 feet. Step 2: 17 ÷ 4 = 4 R1, so 17/4 = 4 1/4 feet.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct? Why is 3/5 equal to 9/15?	**3/5 × 3/3 = 9/15, and 3/3 equals 1, so the amount does not change.**	Multiplying the numerator and denominator by the same number is the same as multiplying by 1 (3/3 = 1), so the value stays the same. Adding numbers does not keep fractions equal.
x2_s0	Set 1: Find the mistake: Kiara simplified 12/18 and got 6/9. She says 6/9 is in simplest form. What did she do wrong?	**6 and 9 still share the factor 3. Dividing the numerator and denominator of 12/18 by the GCF, 6, gives 2/3.**	6/9 is equal to 12/18, but 6 and 9 still have a common factor of 3, so it is not in simplest form. The GCF of 12 and 18 is 6: 12 ÷ 6 = 2 and 18 ÷ 6 = 3, so 12/18 = 2/3.
x1_s1	Set 2: Which explanation is correct? Why does 17/4 equal 4 1/4?	**17 ÷ 4 = 4 R1, which makes 4 wholes with 1 fourth left over.**	Four fourths make 1 whole. 17 fourths make 4 groups of 4 fourths (4 wholes) with 1 fourth left over, so 17/4 = 4 1/4.
x2_s1	Set 2: Find the mistake: Tomás changed 2 3/5 to an improper fraction and got 6/5. What did he do wrong?	**He multiplied 2 × 3. He should multiply 2 × 5 and add 3 to get 13/5.**	2 wholes = 2 × 5 = 10 fifths. Add the 3 fifths: 10 + 3 = 13. So 2 3/5 = 13/5. Tomás multiplied the whole number by the numerator instead of the denominator.

ID	Question	Answer	Why (explain to student)
x1_s2	Set 3: Which explanation is correct? Why is 15/25 NOT in simplest form?	**15 and 25 have a common factor of 5, so 15/25 simplifies to 3/5.**	A fraction is in simplest form only when the GCF of the numerator and denominator is 1. The GCF of 15 and 25 is 5, so 15/25 = 3/5.
x2_s2	Set 3: Find the mistake: Lin says 3/4 = 5/6 because she added 2 to the numerator and 2 to the denominator. What did she do wrong?	**Adding the same number changes the amount. She must multiply or divide both by the same number.**	3/4 = 9/12 and 5/6 = 10/12, so they are not equal. To make an equivalent fraction, multiply or divide the numerator and denominator by the same number.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
f1_s0	Set 1: Fix the sentence: amara asked, is 6/8 the same as 3/4	**Amara asked, "Is 6/8 the same as 3/4?"**	Capitalize the name Amara and the first word of the quote, put quotation marks around her exact words, and put the question mark inside the closing quotation mark.
f2_s0	Set 1: Fix the sentence: The fraction 9/12 is not in simplest form because it's numerator and denominator share the factor 3.	**The fraction 9/12 is not in simplest form because its numerator and denominator share the factor 3.**	Its (no apostrophe) shows ownership: the numerator and denominator belong to the fraction. It's is short for it is.
f3_s0	Set 1: Fix the sentence: Diego needs blue tiles white tiles and gray tiles for the lobby.	**Diego needs blue tiles, white tiles, and gray tiles for the lobby. OR Diego needs blue tiles, white tiles and gray tiles for the lobby.**	Use commas to separate the items in a series: blue tiles, white tiles, and gray tiles.
f4_s0	Set 1: Fix the sentence: the plumbers measures each pipe in feet	**The plumbers measure each pipe in feet. OR The plumber measures each pipe in feet.**	Capitalize the first word. The subject plumbers is plural, so use measure, not measures. End with a period.
f1_s1	Set 2: Fix the sentence: ms. okafor said, the floor plan is ready	**Ms. Okafor said, "The floor plan is ready."**	Capitalize Ms. Okafor and the first word of the quote, put quotation marks around her exact words, and put the period inside the closing quotation mark.
f2_s1	Set 2: Fix the sentence: Its important to divide by the GCF.	**It's important to divide by the GCF. OR It is important to divide by the GCF.**	It's means it is: it is important. Use the apostrophe.
f3_s1	Set 2: Fix the sentence: The fractions 2/4 3/6 and 4/8 are all equal to 1/2.	**The fractions 2/4, 3/6, and 4/8 are all equal to 1/2. OR The fractions 2/4, 3/6 and 4/8 are all equal to 1/2.**	Use commas to separate the items in a series: 2/4, 3/6, and 4/8.
f4_s1	Set 2: Fix the sentence: each of the tiles are one foot wide	**Each of the tiles is one foot wide.**	Capitalize the first word. The subject is each (one tile at a time), so use is, not are. End with a period.

ID	Question	Answer	Why (explain to student)
f1_s2	Set 3: Fix the sentence: darnell asked, how many bundles of rebar came today	**Darnell asked, "How many bundles of rebar came today?"**	Capitalize Darnell and the first word of the quote, put quotation marks around his exact words, and put the question mark inside the closing quotation mark.
f2_s2	Set 3: Fix the sentence: A mixed number shows it's whole number first, then a fraction.	**A mixed number shows its whole number first, then a fraction.**	Its (no apostrophe) shows ownership: the whole number belongs to the mixed number. It's means it is.
f3_s2	Set 3: Fix the sentence: Houston has hot summers heavy rain and busy highways.	**Houston has hot summers, heavy rain, and busy highways. OR Houston has hot summers, heavy rain and busy highways.**	Use commas to separate the items in a series: hot summers, heavy rain, and busy highways.
f4_s2	Set 3: Fix the sentence: the crew and the engineer checks the plans every morning	**The crew and the engineer check the plans every morning.**	Capitalize the first word. Two subjects (the crew and the engineer) need the verb check, not checks. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH, 10 TOTAL)

ID	Question	Answer	Why (explain to student)
match1_s0	Set 1: 2/3 = ?/9	**6**	3 × 3 = 9, so multiply by 3/3: 2 × 3 = 6. 2/3 = 6/9.
match2_s0	Set 1: 12/16 in simplest form	**3/4**	GCF of 12 and 16 is 4: 12 ÷ 4 = 3, 16 ÷ 4 = 4.
match3_s0	Set 1: 9/4 as a mixed number	**2 1/4**	9 ÷ 4 = 2 R1, so 2 1/4.
match4_s0	Set 1: 1 2/3 as an improper fraction	**5/3**	1 × 3 = 3 thirds, plus 2 thirds = 5/3.
match5_s0	Set 1: GCF of 10 and 15	**5**	Factors of 10: 1, 2, 5, 10. Factors of 15: 1, 3, 5, 15. Greatest shared: 5.
match1_s1	Set 2: 3/4 = ?/16	**12**	4 × 4 = 16, so multiply by 4/4: 3 × 4 = 12. 3/4 = 12/16.
match2_s1	Set 2: 10/25 in simplest form	**2/5**	GCF of 10 and 25 is 5: 10 ÷ 5 = 2, 25 ÷ 5 = 5.
match3_s1	Set 2: 7/2 as a mixed number	**3 1/2**	7 ÷ 2 = 3 R1, so 3 1/2.
match4_s1	Set 2: 2 1/4 as an improper fraction	**9/4**	2 × 4 = 8 fourths, plus 1 fourth = 9/4.
match5_s1	Set 2: GCF of 14 and 21	**7**	Factors of 14: 1, 2, 7, 14. Factors of 21: 1, 3, 7, 21. Greatest shared: 7.
match1_s2	Set 3: 4/5 = ?/20	**16**	5 × 4 = 20, so multiply by 4/4: 4 × 4 = 16. 4/5 = 16/20.
match2_s2	Set 3: 6/27 in simplest form	**2/9**	GCF of 6 and 27 is 3: 6 ÷ 3 = 2, 27 ÷ 3 = 9.
match3_s2	Set 3: 11/3 as a mixed number	**3 2/3**	11 ÷ 3 = 3 R2, so 3 2/3.
match4_s2	Set 3: 3 1/2 as an improper fraction	**7/2**	3 × 2 = 6 halves, plus 1 half = 7/2.
match5_s2	Set 3: GCF of 18 and 27	**9**	Factors of 18: 1, 2, 3, 6, 9, 18. Factors of 27: 1, 3, 9, 27. Greatest shared: 9.

BOSS BATTLE: INSPECTION DAY (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: Inspection day! The building has 48 apartments, and 36 of them have their floors tiled. Down on the ground floor, Mr. Nguyen lines up 10 short pieces of pipe end to end. Each piece is 1/4 foot long. Two steps: what fraction of the apartments are NOT tiled yet? Type it in simplest form. Fraction not tiled =	**1/4**	Step 1: 48 − 36 = 12 apartments are not tiled, so 12/48. Step 2: the GCF of 12 and 48 is 12. 12 ÷ 12 = 1 and 48 ÷ 12 = 4, so the answer is 1/4.
b2_s0	Set 1: Inspection day! The building has 48 apartments, and 36 of them have their floors tiled. Down on the ground floor, Mr. Nguyen lines up 10 short pieces of pipe end to end. Each piece is 1/4 foot long. Two steps: how long is the line of pipe? Write it as a mixed number in simplest form.	**2 1/2 feet**	Step 1: 10 pieces of 1/4 foot make 10/4 feet. 10 ÷ 4 = 2 R2, so 2 2/4 feet. Step 2: simplify the fraction part: 2/4 = 1/2, so the answer is 2 1/2 feet.
b1_s1	Set 2: Inspection day! The building has 60 apartments, and 36 of them have been painted. Down on the ground floor, Mr. Nguyen lines up 14 short pieces of pipe end to end. Each piece is 1/4 foot long. Two steps: what fraction of the apartments are NOT painted yet? Type it in simplest form. Fraction not painted =	**2/5**	Step 1: 60 − 36 = 24 apartments are not painted, so 24/60. Step 2: the GCF of 24 and 60 is 12. 24 ÷ 12 = 2 and 60 ÷ 12 = 5, so the answer is 2/5.
b2_s1	Set 2: Inspection day! The building has 60 apartments, and 36 of them have been painted. Down on the ground floor, Mr. Nguyen lines up 14 short pieces of pipe end to end. Each piece is 1/4 foot long. Two steps: how long is the line of pipe? Write it as a mixed number in simplest form.	**3 1/2 feet**	Step 1: 14 pieces of 1/4 foot make 14/4 feet. 14 ÷ 4 = 3 R2, so 3 2/4 feet. Step 2: simplify the fraction part: 2/4 = 1/2, so the answer is 3 1/2 feet.
b1_s2	Set 3: Inspection day! The building has 72 apartments, and 48 of them have their windows in. Down on the ground floor, Mr. Nguyen lines up 18 short pieces of pipe end to end. Each piece is 1/4 foot long. Two steps: what fraction of the apartments do NOT have windows yet? Type it in simplest form. Fraction without windows =	**1/3**	Step 1: 72 − 48 = 24 apartments do not have windows, so 24/72. Step 2: the GCF of 24 and 72 is 24. 24 ÷ 24 = 1 and 72 ÷ 24 = 3, so the answer is 1/3.
b2_s2	Set 3: Inspection day! The building has 72 apartments, and 48 of them have their windows in. Down on the ground floor, Mr. Nguyen lines up 18 short pieces of pipe end to end. Each piece is 1/4 foot long. Two steps: how long is the line of pipe? Write it as a mixed number in simplest form.	**4 1/2 feet**	Step 1: 18 pieces of 1/4 foot make 18/4 feet. 18 ÷ 4 = 4 R2, so 4 2/4 feet. Step 2: simplify the fraction part: 2/4 = 1/2, so the answer is 4 1/2 feet.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	2 of 4 tiles blue: two fractions	2/4 and 1/2 (any equivalent pair is correct, such as 2/4 and 4/8). The drawing should show 4 equal parts with 2 shaded.
2	Which fraction equals 1/2?	3/6 (3 is half of 6).
3	1/3 = ?/6	2 (multiply by 2/2: 1 × 2 = 2, 3 × 2 = 6).
4	Match the words	numerator: the top number of a fraction; denominator: the bottom number of a fraction; equivalent fractions: fractions that name the same amount.
5	Factors of 12 and 18; GCF	12: 1, 2, 3, 4, 6, 12. 18: 1, 2, 3, 6, 9, 18. GCF = 6 (circled).

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	12 of 30 tiles blue	12/30 = 2/5. The GCF of 12 and 30 is 6: 12 ÷ 6 = 2, 30 ÷ 6 = 5.
2	Three fractions equivalent to 2/3	Any three correct, each with its n/n, for example 4/6 (× 2/2), 6/9 (× 3/3), 8/12 (× 4/4).
3	23/6 as a mixed number	3 5/6 (23 ÷ 6 = 3 R5).
4	4 5/8 as an improper fraction	37/8. 4 wholes = 4 × 8 = 32 eighths; 32 + 5 = 37.
5	Writing: is 9/12 in simplest form?	He is wrong. Model: "9/12 is not in simplest form. The greatest common factor of 9 and 12 is 3, so I can divide both by 3 to get 3/4. A fraction is in simplest form only when the greatest common factor of its numerator and denominator is 1." Look for complete sentences, the GCF of 3, the answer 3/4, and both vocabulary words.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	3/4 with denominator 30–40, numerator a multiple of 9	27/36. Multiples of 4 between 30 and 40 are 32 and 36. 3/4 = 24/32 (24 is not a multiple of 9) and 3/4 = 27/36 (27 = 9 × 3), so 27/36 is the only answer.